

Koral Kulacoglu

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EDUCATION

University of Waterloo | Bachelor of Software Engineering (AI Specialization) 3.8/4.0 GPA | Sep 2023 – Apr 2028

- **Awards:** [Hack the North Winner](#) [🔗](#), [USACO Platinum](#), [1st place at Canada-Wide Science Fair + 2nd at EUCYS](#) [🔗](#)
- **Math:** Combinatorics & Optimization, Numerical Computation, Signals & Systems, Statistics, Linear Algebra, Calculus
- **CS:** Advanced Algorithms, Operating Systems, Database Systems, Compilers, Software Design & Testing, User Interfaces

EXPERIENCE

Coinbase | Software Engineer Intern San Francisco, CA | May 2026 - Aug 2026

- Built a **Go risk pipeline** (**Temporal**, **PostgreSQL**) feeding risk parameters across **85+** assets to the [Institutional Financing](#) [🔗](#) margin engine, cutting staleness from **24h to 30min** and reducing liquidation risk on **\$1.3B+** in financed positions.
- Added per-stage **latency tracking** to **cross-margin** endpoints to make oncall alerts actionable, then analyzed staging traffic in **Datadog** to surface the per-entity margin fan-out as a dominant cost (**781ms p95, 4.7×** the next stage).

Optiver | FutureFocus Tech Chicago, IL | Mar 2026

- Engineered a deterministic **TCP-based electronic exchange** in **Python**, utilizing socket programming and unit testing.

BitGo | Software Engineer Intern Palo Alto, CA | Sep 2025 – Dec 2025

- Built a **Rust-based linter** using **Tree-sitter** and **GraphQL** to **lazy-parse** package manifests and prevent vulnerable dependency updates, protecting **320+** BitGo repositories from supply chain attacks.
- Parallelized BitGo's static analysis engine via **shared-state concurrency** (Arc/Mutex), cutting runtime by **95%** (5m to 15s), and refined integration tests by implementing API mocking via **Mockito**.

AtoB | Software Engineer Intern San Francisco, CA | Jan 2025 – Apr 2025

- Created lead data pipelines, including a **Ruby on Rails UTM tracking system** ingesting **500K+** events monthly, and a **Python Selenium UCC-1 web scraper**, collecting **\$150K+** worth of data via **Snowflake**.
- Developed a **greedy semantic matching algorithm** using a priority queue and **Sentence Transformers** to automate PDF-to-SQL mapping in the [Fuel Statement Analyzer](#) [🔗](#), saving **\$2** of manual labor per analysis.

PROJECTS

🐙 FIX Exchange Apr 2026 – Ongoing

- Engineered a **C++ equity exchange** with **FIX 4.2** over **TCP**, price-time priority matching, and **UDP multicast** market data.
- Benchmarked on **AWS EC2 c6i.metal** with **28μs engine latency** (p99) and **27K orders/sec**; automated via **GitHub Actions**.
- Reduced p99 match latency **99.8%** through cache-optimized **absl::btree_map/flat hash map** containers, while eliminating kernel jitter via **pure-spin loops** (`_mm_pause`), **SCHED_FIFO** scheduling, **CPU pinning**, and **SPSC** lock-free queues.
- Built a persistent **SQLite** database with **write-ahead logging** and **async batch-processing** for deterministic state recovery.

🐙 Partcl/HRT Macro Placement Challenge May 2026 – May 2026

- Built a GPU-optimized, pure **PyTorch/NumPy** macro placement engine using **FFT electrostatic global placement** with **Lin-Kernighan k-opt** and **hierarchical Late-Acceptance Hill Climbing**, beating the RePlAce baseline by **30%**.
- Engineered a **bit-exact incremental** reimplement of the proxy cost (within **1e-3** of the oracle) with **delta updates** per macro move, cutting evaluation **2000×** (4s to 2ms), enabling **millions** of Lin-Kernighan/LAHC trial moves under **55min**.

SKILLS

- **Languages:** C/C++, Python, Rust, Go, Ruby, Java, JavaScript, TypeScript, SQL, Shell Script, Assembly, LaTeX
- **Tools:** Linux, Git, Docker, AWS, Kubernetes, CMake, GraphQL, Temporal, TensorFlow, PyTorch, NumPy, Rails, NodeJS